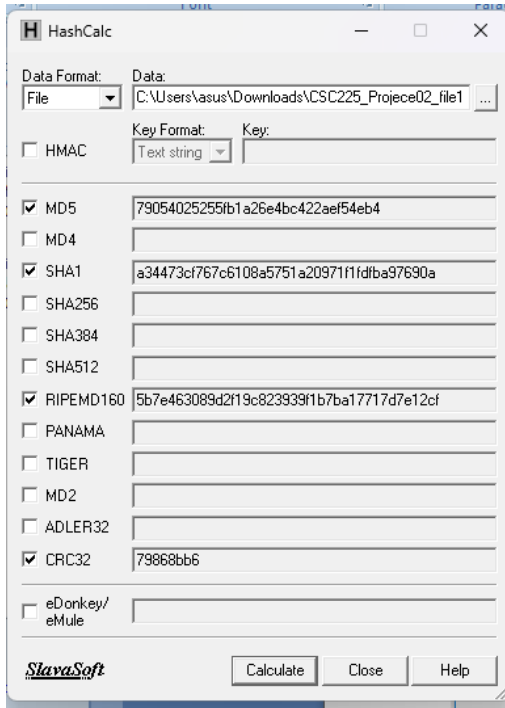
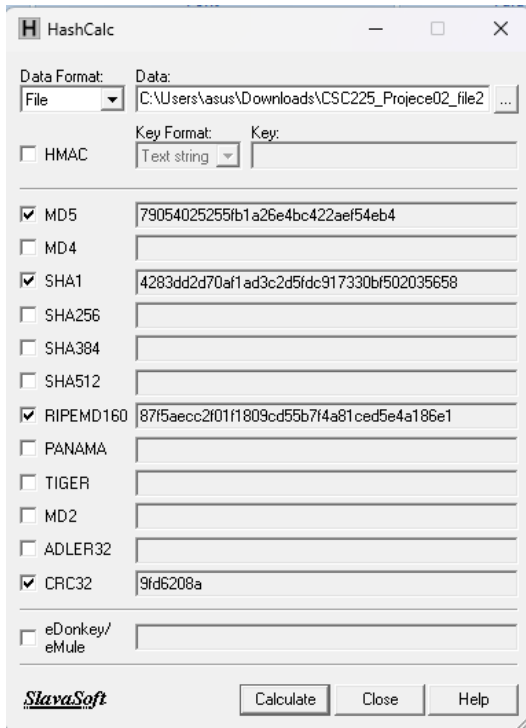
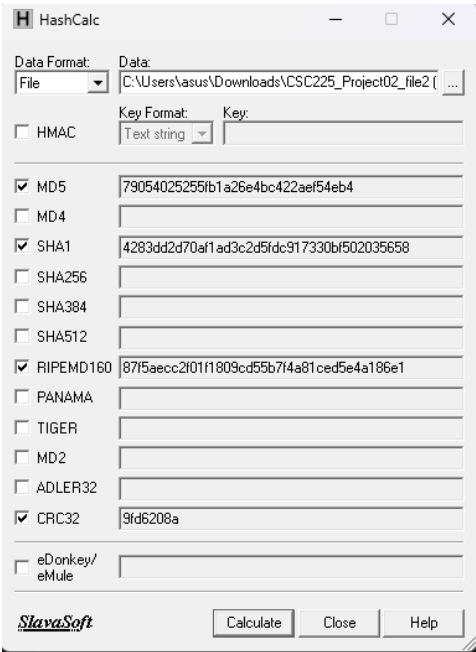


Project 02

Hash for file 1



Hash for file 2



Hash for file after change in name

Hex comparisons for both files

CSC225_Projece02_fil...	CSC225_Projece02_fil...	CSC225_Project02_fil...
D1 31 DD 02 C5 E6 EE C4	69 3D 9A 06 98 AF F9 5C	
2F CA B5 87 12 46 7E AB	40 04 58 3E B8 FB 7F 89	
55 AD 34 06 09 F4 B3 02	83 E4 88 83 25 71 41 5A	
08 51 25 E8 F7 CD C9 9F	D9 1D BD F2 80 37 3C 5B	
D8 82 3E 31 56 34 8F 5B	AE 6D AC D4 36 C9 19 C6	
DD 53 E2 B4 87 DA 03 FD	02 39 63 06 D2 48 CD A0	
E9 9F 33 42 0F 57 7E E8	CE 54 B6 70 80 A8 0D 1E	
C6 98 21 BC B6 A8 83 93	96 F9 65 2B 6F F7 2A 70	
+		

CSC225_Projece02_fil...	CSC225_Projece02_fil...	CSC225_Project02_fil...
D1 31 DD 02 C5 E6 EE C4	69 3D 9A 06 98 AF F9 5C	
2F CA B5 07 12 46 7E AB	40 04 58 3E B8 FB 7F 89	
55 AD 34 06 09 F4 B3 02	83 E4 88 83 25 F1 41 5A	
08 51 25 E8 F7 CD C9 9F	D9 1D BD 72 80 37 3C 5B	
D8 82 3E 31 56 34 8F 5B	AE 6D AC D4 36 C9 19 C6	
DD 53 E2 34 87 DA 03 FD	02 39 63 06 D2 48 CD A0	
E9 9F 33 42 0F 57 7E E8	CE 54 B6 70 80 28 0D 1E	
C6 98 21 BC B6 A8 83 93	96 F9 65 AB 6F F7 2A 70	
+		

5. What differences do you see among the MD5, SHA1, and SHA-256 hashes of file1?

ANS: Although all three hash values were generated from the same file (file1), the MD5, SHA1, and SHA-256 hashes are completely different. This is expected because each algorithm uses a different method and produces a different length of output:- MD5 creates a 128-bit hash.- SHA1 creates a 160-bit hash.- SHA-256 creates a 256-bit hash. Even though the input file is the same, the way each algorithm processes the data is unique, which results in very different hash values.

6. What differences do you see between the MD5 hash of file2 and the MD5 hash of file3?

ANS: File3 is just a slightly edited version of file2 (only one character was changed), yet the MD5 hash of file3 is completely different from that of file2. This shows how sensitive MD5 is--changing even one small part of a file creates a totally new hash. This behavior is known as the "avalanche effect," where a tiny change in input leads to a big change in the output.

7. Compare the MD5 hash of file1 and the MD5 hash of file2:

(1) Are they identical?

ANS: Yes, the MD5 hashes of file1 and file2 are exactly the same.

(2) Why is that?

ANS: Even though file1 and file2 are different at the byte level, they were designed to produce the same MD5 hash. This is an example of what's called an "MD5 collision," where two different files end up having the same MD5 value. These types of collisions show a serious weakness in MD5 and are one of the main reasons why MD5 is no longer considered secure for things like digital signatures or file verification in sensitive applications.